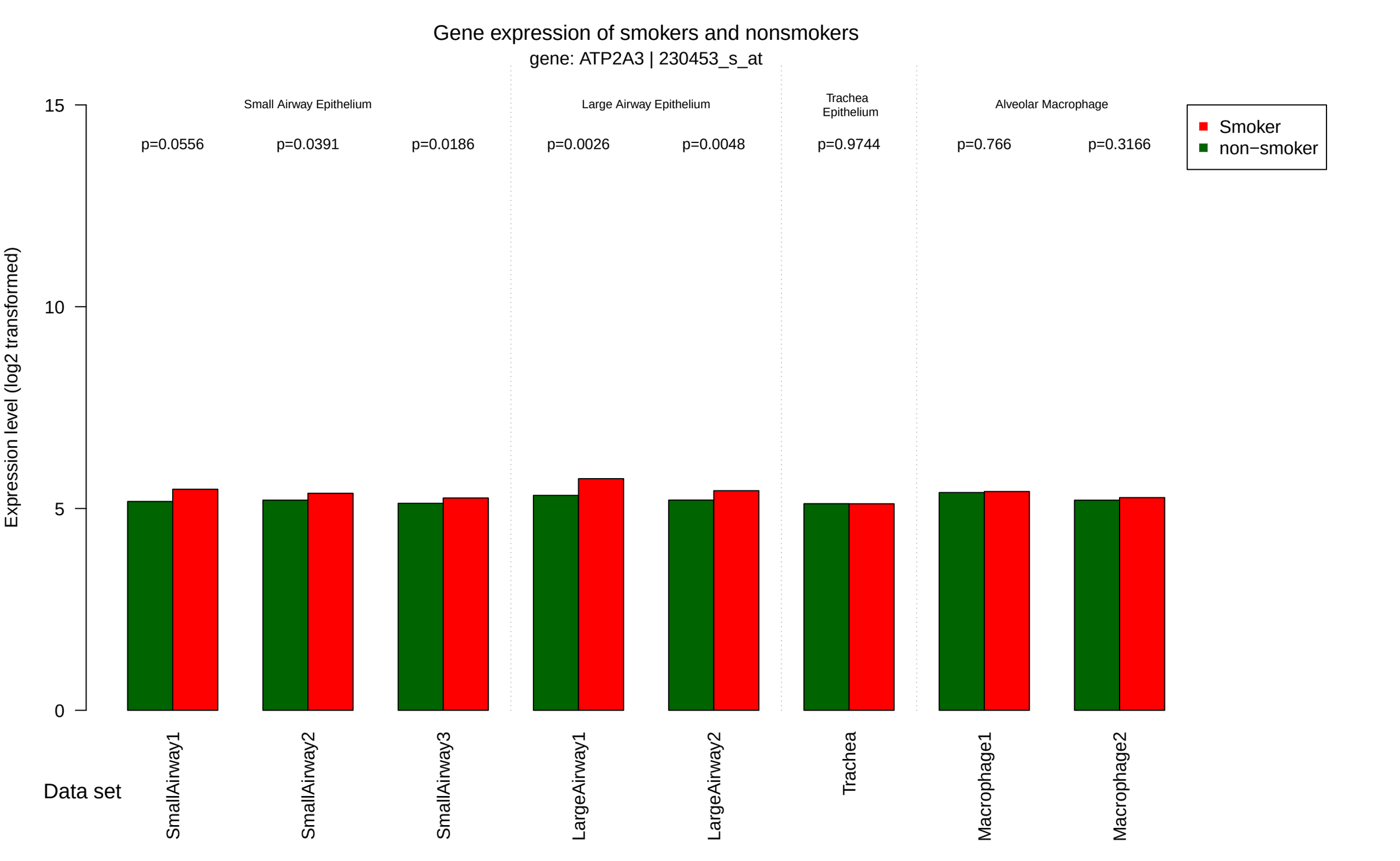
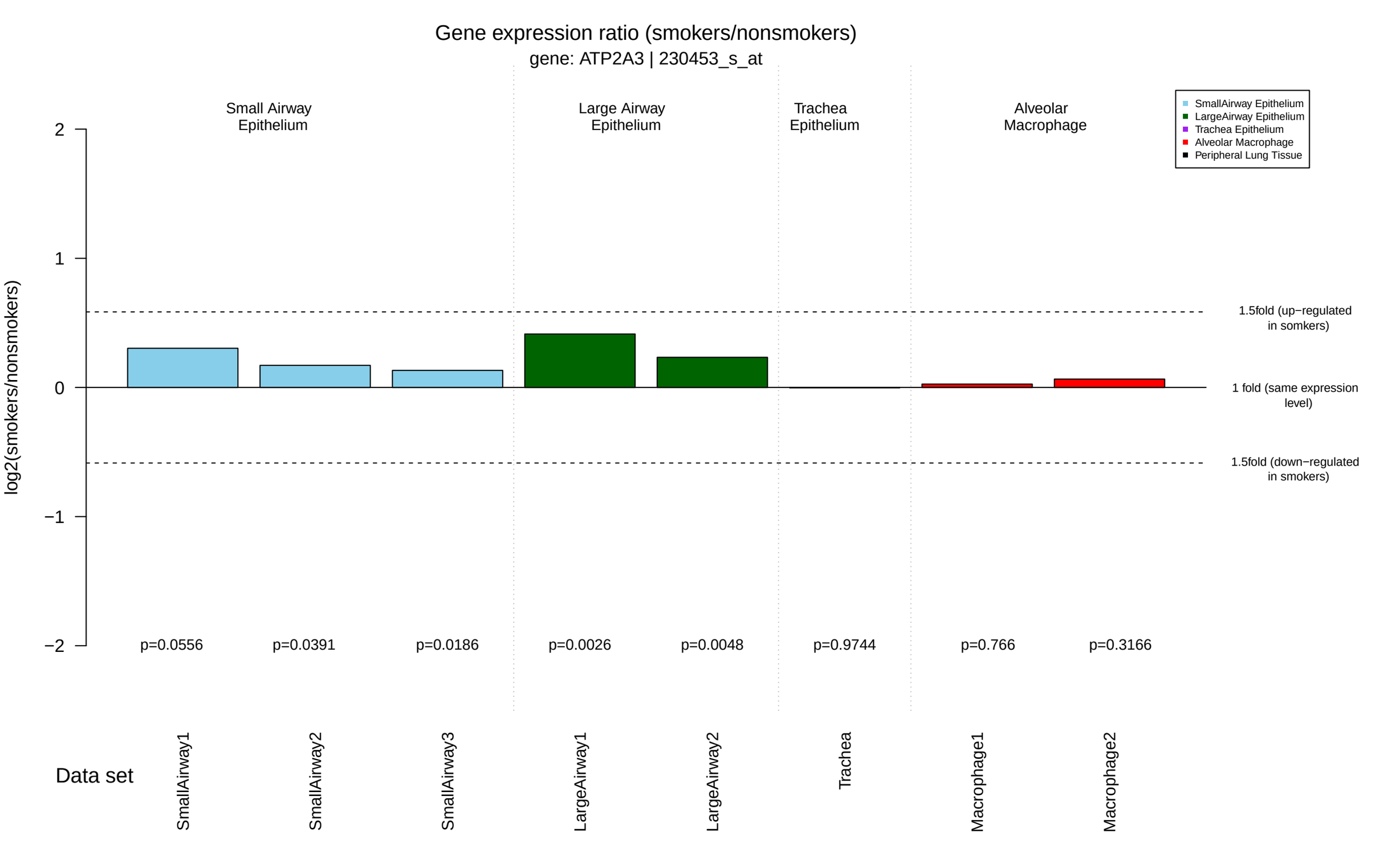
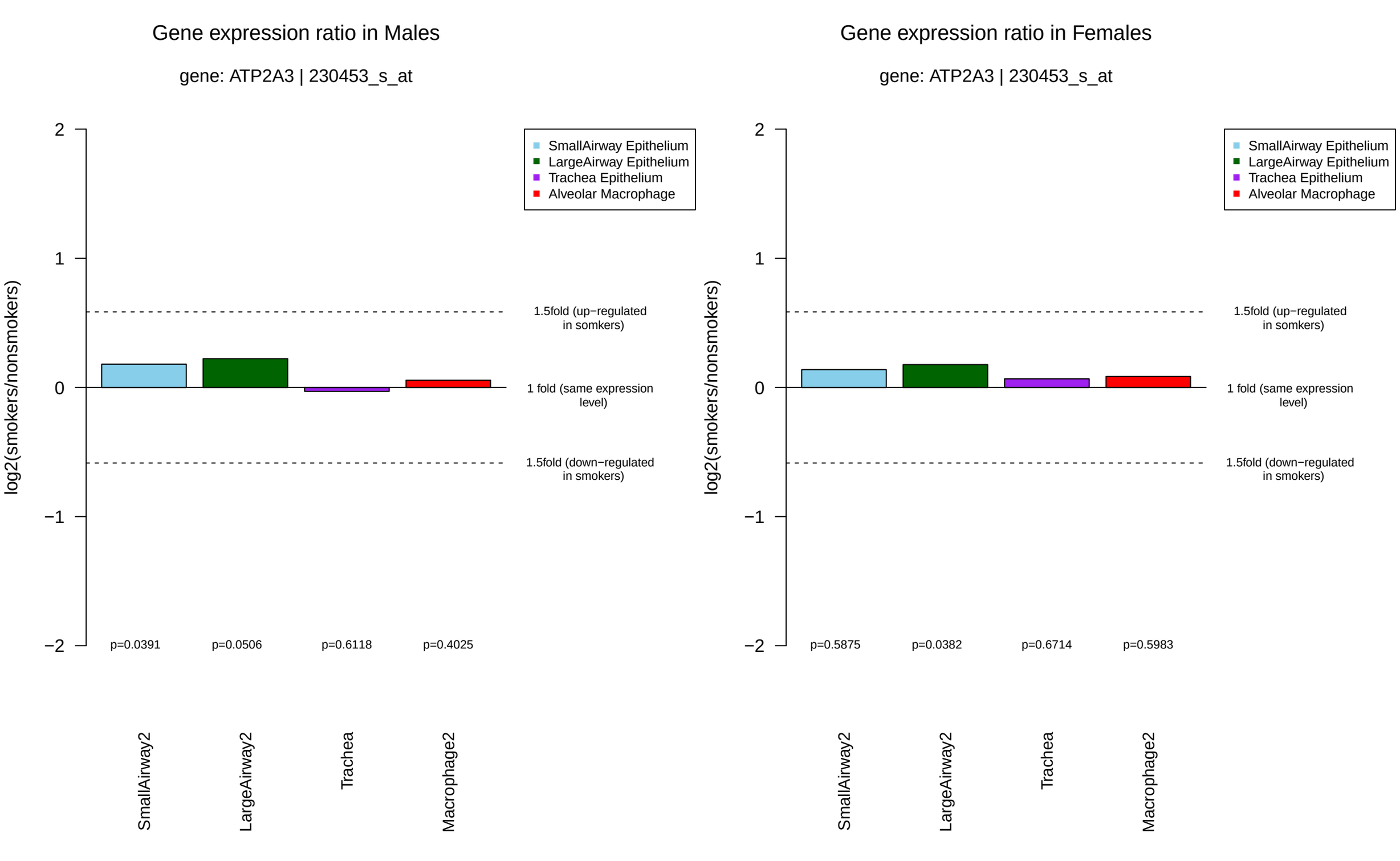
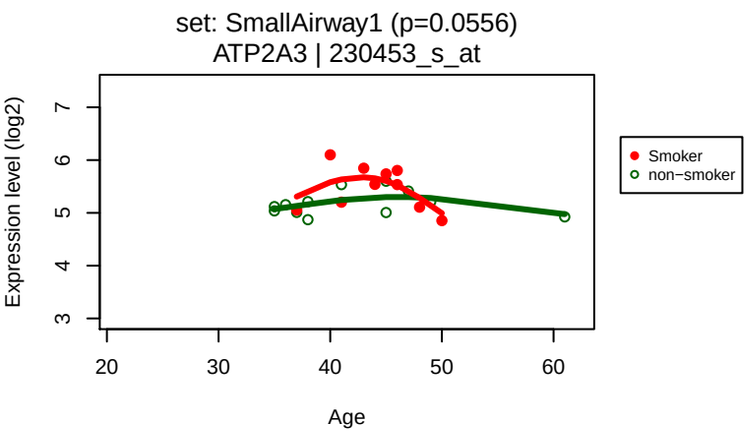
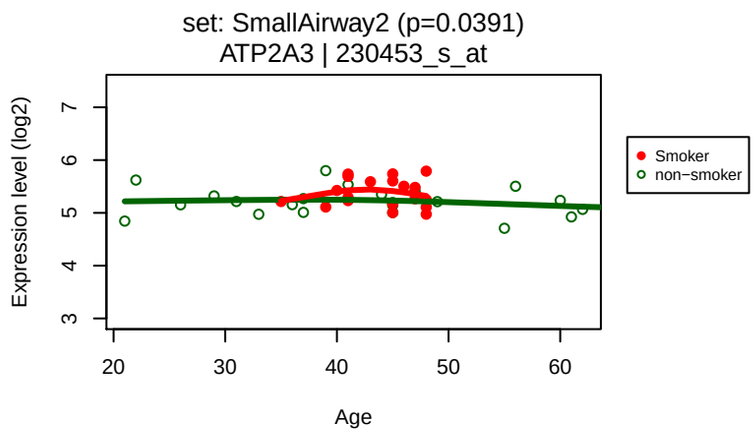
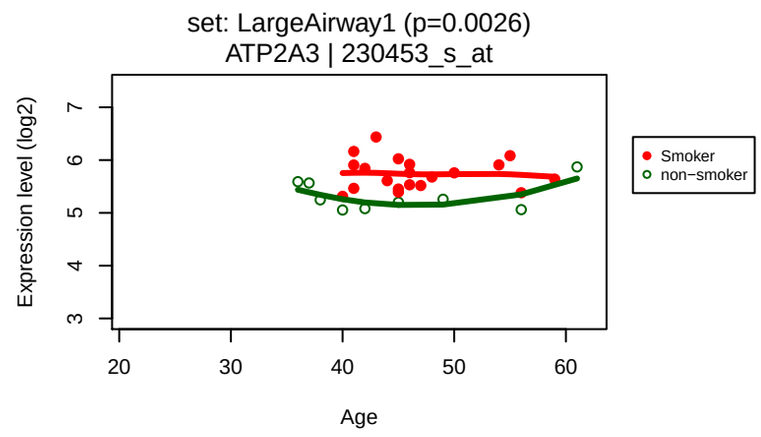

Gene expression of smokers and nonsmokers
gene: ATP2A3 | 230453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0556
p=0.0391
p=0.0186
p=0.0026
p=0.0048
p=0.9744
p=0.766
p=0.3166
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

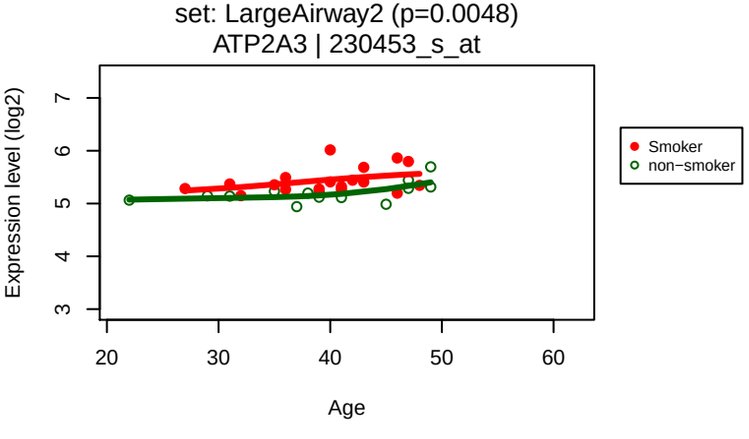

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A3 | 230453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0556
p=0.0391
p=0.0186
p=0.0026
p=0.0048
p=0.9744
p=0.766
p=0.3166
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

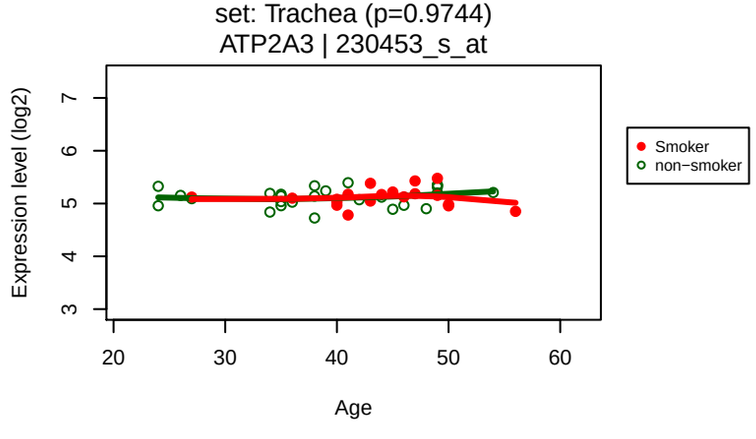

Gene expression ratio in Males
gene: ATP2A3 | 230453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0391
p=0.0506
p=0.6118
p=0.4025
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A3 | 230453_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5875
p=0.0382
p=0.6714
p=0.5983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

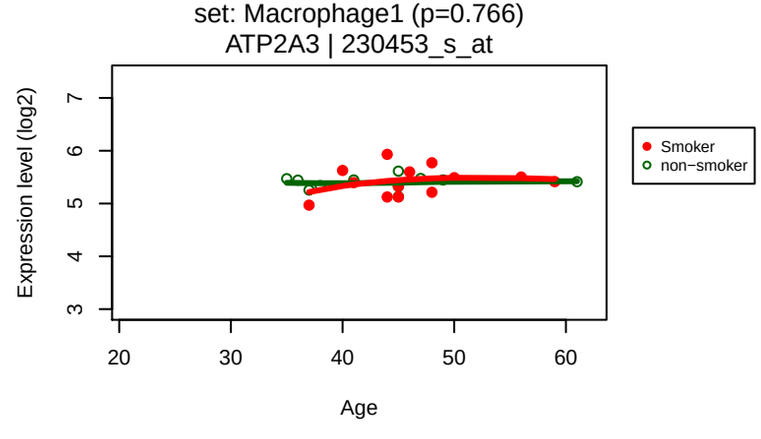

set: SmallAirway1 (p=0.0556)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker

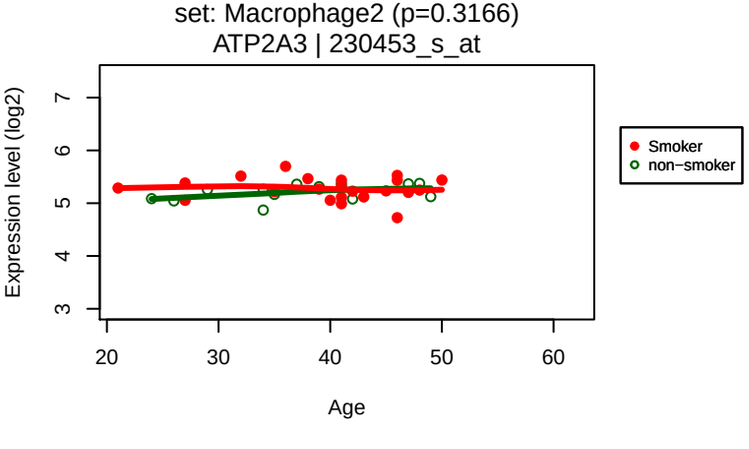

set: SmallAirway2 (p=0.0391)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker

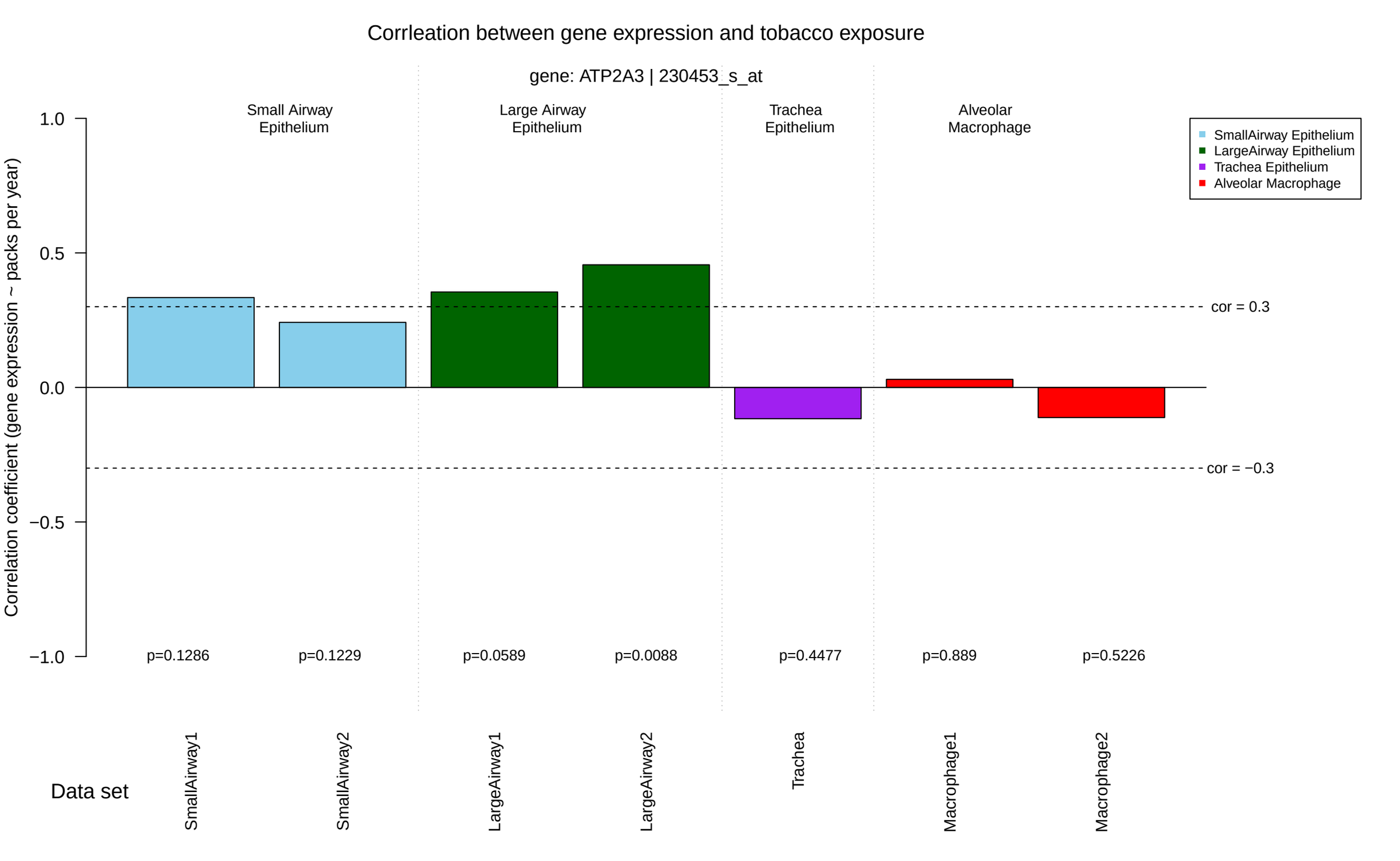

set: LargeAirway1 (p=0.0026)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0048)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9744)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.766)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3166)
ATP2A3 | 230453_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A3 | 230453_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1286
p=0.1229
p=0.0589
p=0.0088
p=0.4477
p=0.889
p=0.5226
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2